Data Preparation

Date	24 October 2026
Team ID	
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	5 Marks

Data preparation refers to the process of cleaning, standardizing, transforming, and organizing data so that it is suitable for analysis and reporting. In a Snowflake project, data preparation involves working with the ingested RAW data, identifying data quality issues, creating STAGING tables, applying required transformations, and preparing reliable data for the CURATED and MART layers.

Activity 1: Identifying and Handling Null Values

Code:

```
SELECT COUNT(*) FROM RAW_ATHLETES WHERE Medal IS NULL;

UPDATE RAW_ATHLETES

SET Medal = 'None'

WHERE Medal IS NULL;
```

Output: [Insert Output Screenshot]

Activity 2: Duplicate Record Handling

Code:

```
CREATE OR REPLACE TABLE STAGING_ATHLETES AS

SELECT DISTINCT * FROM RAW_ATHLETES;
```

Output: [Insert Output Screenshot]

Activity 3: Data Cleaning and Standardization

Code:

```
UPDATE STAGING_ATHLETES

SET NOC = 'GER'

WHERE NOC IN ('FRG', 'GDR');

UPDATE STAGING_ATHLETES

SET NOC = 'RUS'
```

WHERE NOC IN ('URS', 'EUN');

Output: [Insert Output Screenshot]

Activity 4: Data Transformation and Validation

Code:

```
CREATE OR REPLACE VIEW CURATED.MART_MEDALS AS
SELECT
ID as athlete_id,
Name as athlete_name,
Sex as gender,
Age as age,
NOC as noc_code,
Games as games,
Year as game_year,
Season as season,
Sport as sport,
Medal as medal
FROM STAGING_ATHLETES;
```

Output: [Insert Output Screenshot]